

Credit Card Transactions

Data Analysis Case Study using MS SQL Server

**Note: By using stored procedures one could provide fast, secure, reusable analytics with automatic audit logging for production-ready performance.*

Stored Procedure based Analysis

1. Write a stored procedure that accepts a city name as input and returns the top 10 transactions (all columns) for that city ordered by transaction_date descending.

@CityName = 'Delhi' --execute

Results		Messages					
	transaction_id	city	transaction_date	card_type	exp_type	gender	amount
1	13424	Delhi	2015-05-26	Platinum	Entertainment	F	245806
2	13324	Delhi	2015-05-26	Signature	Grocery	F	33068
3	12574	Delhi	2015-05-26	Gold	Entertainment	M	63264
4	9980	Delhi	2015-05-26	Platinum	Entertainment	F	200932
5	9705	Delhi	2015-05-26	Platinum	Bills	F	5133
6	7294	Delhi	2015-05-26	Platinum	Fuel	F	186122
7	5773	Delhi	2015-05-26	Platinum	Food	M	299056
8	1077	Delhi	2015-05-26	Silver	Entertainment	F	41161
9	7063	Delhi	2015-05-25	Signature	Fuel	F	238780
10	3209	Delhi	2015-05-25	Platinum	Bills	F	110862

2. Write a stored procedure that takes a card_type and a year as parameters and returns the total spend per month for that card_type in that year.

exec usp_GetMonthlySpendByCardTypeAndYear @CardType='Gold', @Year=2014; --execute

Results		Messages	
	MonthNumber	MonthName	TotalSpend
1	1	January	54691140
2	2	February	45377610
3	3	March	49734572
4	4	April	51578788
5	5	May	52158856
6	6	June	48104062
7	7	July	46066151
8	8	August	49719139
9	9	September	53155064
10	10	October	50057995
11	11	November	51563543
12	12	December	48481156

3. Write a stored procedure that accepts a minimum and maximum amount range and returns all transactions where the amount falls within that range, along with total count and total spend as output parameters.

```
declare @Count int, @Total decimal(18,2);
```

```
exec usp_GetTransactionsByAmountRange --execute
```

```
@MinAmount = 50000,  
@MaxAmount = 200000,  
@TotalCount = @Count output,  
@TotalSpend = @Total output;
```

```
select @Count as TransactionCount, @Total as TotalSpend;
```

Results

Messages

	transaction_id	city	transaction_date	card_type	exp_type	gender	amount
1	25547	Surat	2015-05-26	Platinum	Travel	M	72158
2	24809	Surat	2015-05-26	Silver	Fuel	F	143502
3	24598	Hyderabad	2015-05-26	Gold	Grocery	F	113730
4	23197	Lucknow	2015-05-26	Signature	Fuel	M	52278
5	22479	Surat	2015-05-26	Signature	Entertainment	F	178800
6	22448	Kolkata	2015-05-26	Platinum	Entertainment	M	69660
7	22341	Hyderabad	2015-05-26	Platinum	Entertainment	M	171338
8	21610	Hyderabad	2015-05-26	Gold	Grocery	F	176677
9	20021	Jaipur	2015-05-26	Platinum	Entertainment	F	177359
10	12616	Bengaluru	2015-05-26	Gold	Fuel	F	166997
11	12574	Delhi	2015-05-26	Gold	Entertainment	M	63264
12	11566	Greater Mumbai	2015-05-26	Silver	Entertainment	M	180339
12	7204	Delhi	2015-05-26	Platinum	Fuel	F	186122

	TransactionCount	TotalSpend
1	12850	1612615258.00

4. Write a stored procedure that takes an expense type as input and returns, for each city, the percentage contribution of that city's spends for that expense type to the total spend of that expense type.

```
exec usp_ExpenseTypeCityContribution @ExpenseType = 'Entertainment'; --execute
```

Results		Messages		
	city	CitySpend	total_expense_spend	percentage_contribution
1	Ahmedabad	97502266	726437536	13.42
2	Bengaluru	96061083	726437536	13.22
3	Greater Mumbai	95596729	726437536	13.16
4	Delhi	88282990	726437536	12.15
5	Surat	24480461	726437536	3.37
6	Kolkata	21839330	726437536	3.01
7	Kanpur	21446323	726437536	2.95
8	Lucknow	20861099	726437536	2.87
9	Jaipur	20834998	726437536	2.87
10	Chennai	20316575	726437536	2.80
11	Pune	18353730	726437536	2.53
12	Hyderabad	17528716	726437536	2.41
13	Nokha	1084780	726437536	0.15
14	Bikaner	915159	726437536	0.13
15	Imphal	968242	726437536	0.13
16	Mandvi	925419	726437536	0.13
17	Naidupet	849513	726437536	0.12
18	Sardhana	873193	726437536	0.12
19	Tandur	881892	726437536	0.12